ID5059 Coursework 2: Driver Scheduling for Ride-Hailing Service

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Instructions:

- **Deadlines:** Group Contract: 21:00 on Thursday 26th March, 2026. Final Report: 21:00 on Thursday 16th April, 2026.
- **Credit:** 50% of coursework mark, 20% of final module mark.

Please note that **you are expected to have read and understood all the information in this specification** and any accompanying documents at least a week before the both interim and final deadlines. You must contact the lecturers regarding any queries well in advance of the deadlines. As usual, MMS is the definitive source for deadline and credit details.

Lateness

The standard Computer Science penalty for late submission applies (Scheme A: 1 mark per 24 hour period, or part thereof). Importantly, this applies to both deadlines for the group contract and the final report deadline. In the case where a penalty is incurred for both deadlines, then the greater of the two penalities will apply.

Good Academic Practice

As usual, you are reminded to follow the relevant guidelines on good academic practice. See also the following page of the CS student handbook.

Project Brief

For this assignment, you will, along with your group, play the role of a data analytics team working for a ride-hailing platform that is preparing to launch operations in New York City, particularly the borough of *Manhattan*.

The Operations Manager wishes to minimise passenger waiting times while avoiding an excess of idle drivers. Making these decisions requires reliable predictions of future ride demand.

Your company has no in-house historical data of its own. As a surrogate, you will use publicly available taxi trip records for New York City, provided monthly from 2024.

Your primary task is to build predictive models for **ride demand** for the Manhattan area. These predictions should support recommendations about **driver deployment over time and space**. You should also model **fare-related quantities** (such as average fare or total revenue per hour) in order to explore which locations and times are likely to be most profitable for drivers.

You must present something actionable to the Operations Manager, who is not a data scientist. You should therefore focus on clear communication of your methods, results, and recommendations.

Learning objectives

On successful completion of this assignment you should be able to

- investigate and clean a complex real-world dataset appropriately for the task at hand;
- engineer features for the given prediction task;
- fit and evaluate multiple machine learning models;
- interpret results and communicate recommendations to a non-technical stakeholder.

Group Work

For this assignment, we have asked you to work in groups and there are many benefits to this. You all come from different backgrounds and working in groups allows you to learn from the strengths of your peers. Working in groups is an essential part of professional life and an important skill to develop. Having to explain concepts and ideas clearly to others helps to strengthen your understanding.

If you have any serious reasons as to why you should NOT be in the same group as someone else please contact a Student Welfare Officer as soon as possible, and in any event before Wednesday 11th March, by which time I would like to have groups formed. If you are unable or unwilling to raise a concern before groups are allocated, you can still contact a Student Welfare Officer within one week of the groups having been announced. Note that in this case it may be much harder to make group changes.

Please communicate with your team via official university platforms (i.e. email or teams). Alternative group communication channels should be used only if every group member is aware of and agrees to them.

Group Contract

Before starting work on a group project you should agree how you are going to work within the group. It can also be helpful to discuss how you normally work and communicate to set everyone's expectations.

As a group, develop, agree and sign a group work contract. This should help things run more smoothly. There is a **deadline of Thursday 26th March 2026, 9pm to complete the group contract** and upload it to MMS.

Examples of group contracts can be found here and here.

Project Requirements

We leave the goal and scope deliberately open-ended to allow you to demonstrate creativity and judgement as would be expected in a real-world project. However, there are some *mandatory requirements* that you must meet:

- Your goal is to create a model for predicting *ride-demand* and *expected fare/revenue* for Manhattan ride-hailing services based on the provided NYC Taxi data. You can present this in any manner you think useful. For example, you could predict a typical week, or a typical week in each season (if you see a large fluctuation between summer and winter for example).
- For each target variable (demand and revenue), you should fit and evaluate at least three ML models, including a neural network, and carefully fine-tune a final model on each target variable to make predictions.
- You may use either R or Python (or both) for your analysis. Specifically, you can use different languages for different parts of the project as long as this is clearly documented.
- You must create a report, submit your code and a clearly labeled `predictions.csv` file. There will be a further group presentation during week 12.

Data

You will use (yellow) NYC Taxi record data from 'NYC Taxi & Limousine Commission'.

Some key variables (from the official data dictionary) include:

- `tpep_pickup_datetime`: the date/time when the meter was engaged (pickup time).
- `PULocationID`: the TLC Taxi Zone where the pickup occurred.
- `trip_distance`, `payment_type`, `total_amount`, etc. are somewhat self-explanatory.

Full data dictionary found here.

The TLC webpage holds data for many years. You should only use data from **2024** as per the files downloaded in the 'data' folder provided alongside this specification.

Data exploration and preprocessing

Notice that the data is structured by time and zone/location. You should decide on how to best use this structure to create features for your models.

Some hints and tips:

- load the trip records and zone lookup data;
- clean the data (handle missing/invalid datetimes, etc.);
- filter to Manhattan only pickup zones (the map of Manhattan zones is given);
- engineer time features (e.g. hour-of-day, day-of-week, month, weekend/weekday);
- think about including lag features in at least one of your models.

Your group should submit only one set of data exploration. All trained models must use the same cleaned data, or a subset of it.

Model Selection and Development

You should develop at least three models per target variable. **At least one of the developed models for each task should be a neural network.** Besides that requirement, you are free to choose any machine learning models that you like. You must appropriately evaluate the models, and select a model which you will recommend to your manager.

Predictions

Once you have developed your final model, you should make predictions for the test data. You should produce a `predictions.csv` file containing your predictions for demand and revenue for drivers labeled in a digestible format.

The test loss of the submitted predictions should be evaluated but do not form part of the assessment criteria. The purpose is only to sanity-check that the model is reasonable or to help identify problems with the methodology.

> ! Important
>
> You are **not** assessed on achieving the best possible predictive accuracy. You are assessed on process, justification, and interpretation.

Report (for the Operations Manager)

Your report should detail your data cleaning and preprocessing, discuss the at least six models attempted and justfiy your choice of a specific final prediction model for use with new instances. It should also include layman's insights into the relative importance of different features, how they impact predictions and business oriented recommendations.

You could consider some of the following questions:

1. **How well can we predict hourly demand per zone with the given data?**
   Include your chosen metric(s), plus an intuitive explanation (e.g., "typical error is about X rides per hour per zone which increases/decreases in zones Y during times Z").
2. **What drives demand?**
   Identify the most important features (time-of-day effects, weekday/weekend, recent demand lags, etc.) and explain them.
3. **Driver planning recommendation**
   Provide a recommended staffing policy (even if simple), and discuss limitations/risks (e.g., special events, weather not included, etc.).
4. **Revenue insights**
   Identify the most profitable zones and times, and discuss how drivers might use this information to plan their working hours.

Brevity is expected. Your client is busy. It is recommended that the report **does not exceed 6 pages** in length.

Individual Statement

You are asked to submit an individual statement explaining your contribution and assessing how the effort of the project was distributed across the group members. To assess how the effort of the group is distributed you should specify the percentage of work done by each group member. For example, 5 group members with equally distributed effort, you should indicate they have all done 20% of the effort. If two or more group members indicate that there was a significant disparity in effort and can provide evidence such as git logs or document histories, then marks within the group may be adjusted at the discretion of the markers.

Presentation

During week 12, you will give a group presentation to the lecturers (and perhaps a small subset of your classmates in the audience). You should prepare a presentation that does not exceed 10 minutes in length and each member of the group should speak for at least 1 minute of this time.

You should act as the lecturers are the Operations Managers and that we have read your report in advance. Your goal is therefore to summarize your methodology and present any key findings. That is, *do not waste time presenting the project goal*; we know the goal, as we gave you the task.

You should also be prepared to answer questions about your work and justify your decisions.

You will be asked to upload your slides to MMS ASAP after the presentation.

Submission Instructions

Submit a single `.zip` containing:

- The code of your solution, in a notebook (e.g. Jupyter (.ipynb), Rmarkdown(.rmd) or Quarto(.qmd)), with markdown annotations. Each group member submits the same code.
- An HTML or PDF version of your notebook including all output and markdown. Each group member submits the same document.
- A clear and concise summary report describing your models, the performance of your model and your results, recommendations and conclusions, in a PDF file. Each group member submits the same report.
- A text file predictions.csv containing the predictions of your model on your test data. Each group member submits the same file.
- A brief statement of your individual contribution and assessment of distribution of effort within the group (maximum 1 page). Each group member submits a separate statement.

You are reminded of the importance of re-downloading and checking immediately after submission.

Hints

- The specification is deliberately open-ended. Where anything is unclear, you may choose a reasonable interpretation and document it.
- Keep your pipeline reproducible: a reader should be able to run your notebook end-to-end without manual steps.

- You do not need to use every variable in the raw trip record data. Focus on what supports your prediction and explanation.

Assessment Criteria

This assignment will be marked on the standard 20-point scale using the following example mark descriptors; see also the standard CS mark descriptors:

- **0**: No material submitted or no evidence of any attempt to complete the work.
- **1-3**: Little evidence of any significant attempt to complete the work. A very poor presentation of the project.
- **4-6**: No substantial relevant material submitted, or no evidence of significant progress on any of the basic ML process elements. A poor presentation of the project.
- **7-10**: reasonable attempt at most of the basic ML process elements including summary report, perhaps major flaws in the process or limited explanation and analysis justifying choices. A reasonable presentation of some elements of the project, but perhaps with significant flaws in presentation or explanation.
- **11-13**: A competent attempt at most of the basic ML process elements and summary report, poor presentation of code or summary report, explanation of choices and results unclear or shows confusion. Competent presentation of some elements of the project.
- **14-16**: A clearly presented and explained notebook and summary report, demonstrating significant attempts at all of the basic ML process elements, and success in most elements with adequate explanation of choices and results. Component presentation of all elements of the project.
- **17-18**: An excellent and clearly presented notebook and summary report, including appropriate recommendations for data preprocessing, that demonstrates successful implementation of all the basic ML process elements with well explained decisions and analysis of results. A good presentation of all elements of the project, with significant insight into data preprocessing, and important features in the success of the model.
- **19-20**: An exceptionally clearly presented notebook and summary report, demonstrating extensive successful implementation of all the basic ML process elements, very well explained decisions and analysis of results. Assignment conveys significant insight into data preprocessing, and important features in the success of the model. A very good presentation of all elements of the project, with significant insight into data preprocessing, and important features in the performance of the models.